

```
from google.colab import drive
drive.mount('/content/drive')

DATASET_PATH = "/content/drive/MyDrive/data/archive"
```

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).

```
import os

for root, dirs, files in os.walk('/content/drive/MyDrive'):
    if 'metadata.csv' in files:
        print(os.path.join(root, 'metadata.csv'))
        break
```

/content/drive/MyDrive/data/archive/metadata.csv

```
import pandas as pd
df = pd.read_csv("/content/drive/MyDrive/data/archive/metadata.csv")
for alg in sorted(df["algorithm"].unique()):
    print(f"\nAlgorithm{alg}")
    print(df[df["algorithm"]==alg]["exp_name"].unique()[:20])
```

```
Algorithm0
['cv4_fastspeech2' 'ek1_tacotron2' 'ljspeech_deepvoice3'
 'ljspeech_fast_pitch' 'ljspeech_fastspeech2' 'ljspeech_glow-tts'
 'ljspeech_tacotron2-DCA' 'ljspeech_tacotron2-DDC'
 'ljspeech_tacotron2-DDC_ph' 'ljspeech_vits' 'sam_tacotron-DDC'
 'silero_lj_16khz' 'silero_lj_16khz_enhanced'
 'tf_ljspeech_FASTSPEECH2_MB-MELGAN' 'tf_ljspeech_FASTSPEECH2_MELGAN'
 'tf_ljspeech_FASTSPEECH2_MELGAN-STFT' 'tf_ljspeech_FASTSPEECH_MB-MELGAN'
 'tf_ljspeech_FASTSPEECH_MELGAN' 'tf_ljspeech_FASTSPEECH_MELGAN-STFT'
 'tf_ljspeech_TACOTRON_MELGAN']

Algorithm1
['cv4_fastspeech2' 'ek1_tacotron2' 'ljspeech_deepvoice3'
 'ljspeech_fast_pitch' 'ljspeech_fastspeech2' 'ljspeech_glow-tts'
 'ljspeech_tacotron2-DCA' 'ljspeech_tacotron2-DDC'
 'ljspeech_tacotron2-DDC_ph' 'ljspeech_vits' 'sam_tacotron-DDC'
 'silero_lj_16khz' 'silero_lj_16khz_enhanced'
 'tf_ljspeech_FASTSPEECH2_MB-MELGAN' 'tf_ljspeech_FASTSPEECH2_MELGAN'
 'tf_ljspeech_FASTSPEECH2_MELGAN-STFT' 'tf_ljspeech_FASTSPEECH_MB-MELGAN'
 'tf_ljspeech_FASTSPEECH_MELGAN' 'tf_ljspeech_FASTSPEECH_MELGAN-STFT'
 'tf_ljspeech_TACOTRON_MELGAN']

Algorithm2
['cv4_fastspeech2' 'ek1_tacotron2' 'ljspeech_deepvoice3'
 'ljspeech_fast_pitch' 'ljspeech_fastspeech2' 'ljspeech_glow-tts'
 'ljspeech_tacotron2-DCA' 'ljspeech_tacotron2-DDC'
 'ljspeech_tacotron2-DDC_ph' 'ljspeech_vits' 'sam_tacotron-DDC'
 'silero_lj_16khz' 'silero_lj_16khz_enhanced'
 'tf_ljspeech_FASTSPEECH2_MB-MELGAN' 'tf_ljspeech_FASTSPEECH2_MELGAN'
 'tf_ljspeech_FASTSPEECH2_MELGAN-STFT' 'tf_ljspeech_FASTSPEECH_MB-MELGAN'
 'tf_ljspeech_FASTSPEECH_MELGAN' 'tf_ljspeech_FASTSPEECH_MELGAN-STFT'
 'tf_ljspeech_TACOTRON_MELGAN']

Algorithm3
['ljspeech_tacotron2-DDC' 'ljspeech_tacotron2-DDC_ph'
 'ljspeech_tacotron2-DCA' 'tf_ljspeech_TACOTRON_MELGAN'
 'tf_ljspeech_TACOTRON_MELGAN-STFT' 'sam_tacotron-DDC' 'ek1_tacotron2'
 'silero_lj_16khz' 'silero_lj_16khz_enhanced' 'ljspeech_glow-tts'
 'vctk_sc-glow-tts_speakerrandom' 'ljspeech_vits'
 'vctk_vits_speakerrandom' 'tf_ljspeech_FASTSPEECH_MELGAN'
 'tf_ljspeech_FASTSPEECH_MELGAN-STFT' 'tf_ljspeech_FASTSPEECH_MB-MELGAN'
 'ljspeech_fastspeech2' 'cv4_fastspeech2' 'tf_ljspeech_FASTSPEECH2_MELGAN'
 'tf_ljspeech_FASTSPEECH2_MELGAN-STFT']

Algorithm4
['cv4_fastspeech2' 'ek1_tacotron2' 'ljspeech_deepvoice3'
 'ljspeech_fast_pitch' 'ljspeech_fastspeech2' 'ljspeech_glow-tts'
 'ljspeech_tacotron2-DCA' 'ljspeech_tacotron2-DDC'
 'ljspeech_tacotron2-DDC_ph' 'ljspeech_vits' 'sam_tacotron-DDC'
 'silero_lj_16khz' 'silero_lj_16khz_enhanced'
 'tf_ljspeech_FASTSPEECH2_MB-MELGAN' 'tf_ljspeech_FASTSPEECH2_MELGAN'
 'tf_ljspeech_FASTSPEECH2_MELGAN-STFT' 'tf_ljspeech_FASTSPEECH_MB-MELGAN'
 'tf_ljspeech_FASTSPEECH_MELGAN' 'tf_ljspeech_FASTSPEECH_MELGAN-STFT']
```

```
'tf_ljspeech_TACOTRON_MELGAN']
```

```
Algorithm5
```

```
print("Total samples:", len(df))
```

```
Total samples: 34850
```

```
print("unique TTS systems:", df["exp_name"].nunique())
```

```
unique TTS systems: 25
```

```
print("unique speakers:", df["speaker_id"].nunique())
```

```
unique speakers: 219
```

```
print("\nsamples per TTS system:")
print(df["exp_name"].value_counts())
```

```
samples per TTS system:
exp_name
cv4_fastspeech2          1394
ek1_tacotron2           1394
ljspeech_deepvoice3      1394
ljspeech_fast_pitch      1394
ljspeech_fastspeech2     1394
ljspeech_glow-tts        1394
ljspeech_tacotron2-DCA   1394
ljspeech_tacotron2-DDC   1394
ljspeech_tacotron2-DDC_ph 1394
ljspeech_vits            1394
sam_tacotron-DDC        1394
silero_lj_16khz         1394
silero_lj_16khz_enhanced 1394
tf_ljspeech_FASTSPEECH2_MB-MELGAN 1394
tf_ljspeech_FASTSPEECH2_MELGAN 1394
tf_ljspeech_FASTSPEECH2_MELGAN-STFT 1394
tf_ljspeech_FASTSPEECH_MB-MELGAN 1394
tf_ljspeech_FASTSPEECH_MELGAN 1394
tf_ljspeech_FASTSPEECH_MELGAN-STFT 1394
tf_ljspeech_TACOTRON_MELGAN 1394
tf_ljspeech_TACOTRON_MELGAN-STFT 1394
vctk_deepvoice3_speakerrandom 1394
vctk_fast_pitch_speakerrandom 1394
vctk_sc-glow-tts_speakerrandom 1394
vctk_vits_speakerrandom 1394
Name: count, dtype: int64
```

```
print(df.groupby("exp_name")["speaker_id"].nunique())
```

```
exp_name
cv4_fastspeech2          1
ek1_tacotron2           1
ljspeech_deepvoice3      1
ljspeech_fast_pitch      1
ljspeech_fastspeech2     1
ljspeech_glow-tts        1
ljspeech_tacotron2-DCA   1
ljspeech_tacotron2-DDC   1
ljspeech_tacotron2-DDC_ph 1
ljspeech_vits            1
sam_tacotron-DDC        1
silero_lj_16khz         1
silero_lj_16khz_enhanced 1
tf_ljspeech_FASTSPEECH2_MB-MELGAN 1
tf_ljspeech_FASTSPEECH2_MELGAN 1
tf_ljspeech_FASTSPEECH2_MELGAN-STFT 1
tf_ljspeech_FASTSPEECH_MB-MELGAN 1
tf_ljspeech_FASTSPEECH_MELGAN 1
tf_ljspeech_FASTSPEECH_MELGAN-STFT 1
tf_ljspeech_TACOTRON_MELGAN 1
tf_ljspeech_TACOTRON_MELGAN-STFT 1
vctk_deepvoice3_speakerrandom 107
vctk_fast_pitch_speakerrandom 108
vctk_sc-glow-tts_speakerrandom 108
vctk_vits_speakerrandom 108
Name: speaker_id, dtype: int64
```

```
systems = ["ljspeech vits", "ljspeech fastspeech2", "ljspeech glow-tts"]
```

```
for system in systems:
    row = df[df["exp_name"]==system].iloc[0]
    print("\nsystem:",system)
    print("speaker:",row["speaker_id"])
    print("Audio Path:",row["audio_path"])
```

```
system: ljspeech_vits
speaker: lj
Audio Path: /kaggle/input/kothac1112131415/tmp/Dataset/ljspeech_vits/1059df5984efbd52824922d66db379ff.wav

system: ljspeech_fastspeech2
speaker: lj
Audio Path: /kaggle/input/kothaf01/tmp/Dataset/ljspeech_fastspeech2/7385c5f867ef1df649da476859b8f224.wav

system: ljspeech_glow-tts
speaker: lj
Audio Path: /kaggle/input/kothac678910/tmp/Dataset/ljspeech_glow-tts/b7d53f6c8301665c8c1ae0d647208ecf.wav
```

```
import matplotlib.pyplot as plt
import librosa.display
import numpy as np

import librosa
y,sr = librosa.load("/content/drive/MyDrive/data/archive/wavs/ljspeech_vits/1059df5984efbd52824922d66db379ff.wav")
print("sampling rate:",sr)
print("duration:",len(y)/sr)
print("shape:",y.shape)
plt.figure(figsize=(12,3))
plt.plot(y)
plt.title(system)
plt.xlabel("sample index")
plt.ylabel("amplitude")
plt.show()
D = librosa.amplitude_to_db(np.abs(librosa.stft(y)),ref = np.max)
plt.figure(figsize=(12,5))
librosa.display.specshow(D,sr=sr,x_axis='time',y_axis='hz')
plt.colorbar()
plt.title("spectrogram")
plt.show()

mel = librosa.feature.melspectrogram(y=y,sr=sr,n_mels=128)
log_mel = librosa.power_to_db(mel)
plt.figure(figsize=(12,5))
librosa.display.specshow(log_mel,sr=sr,x_axis='time',y_axis='mel')
plt.colorbar()
plt.title("log-mel spectrogram")
plt.show()

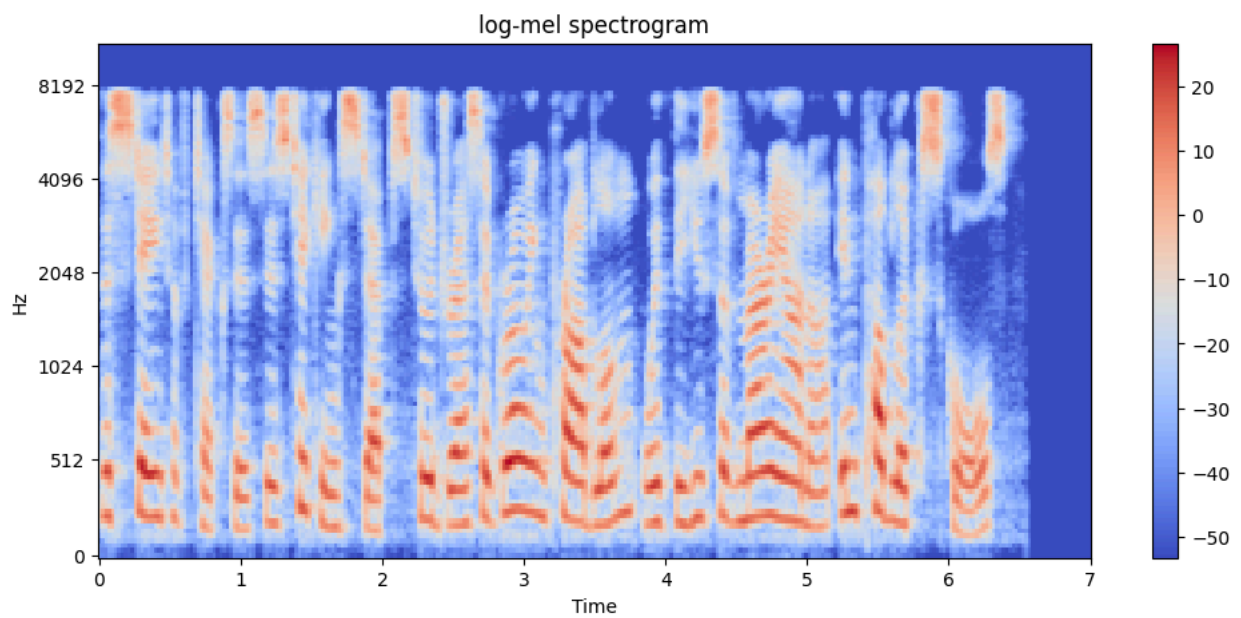
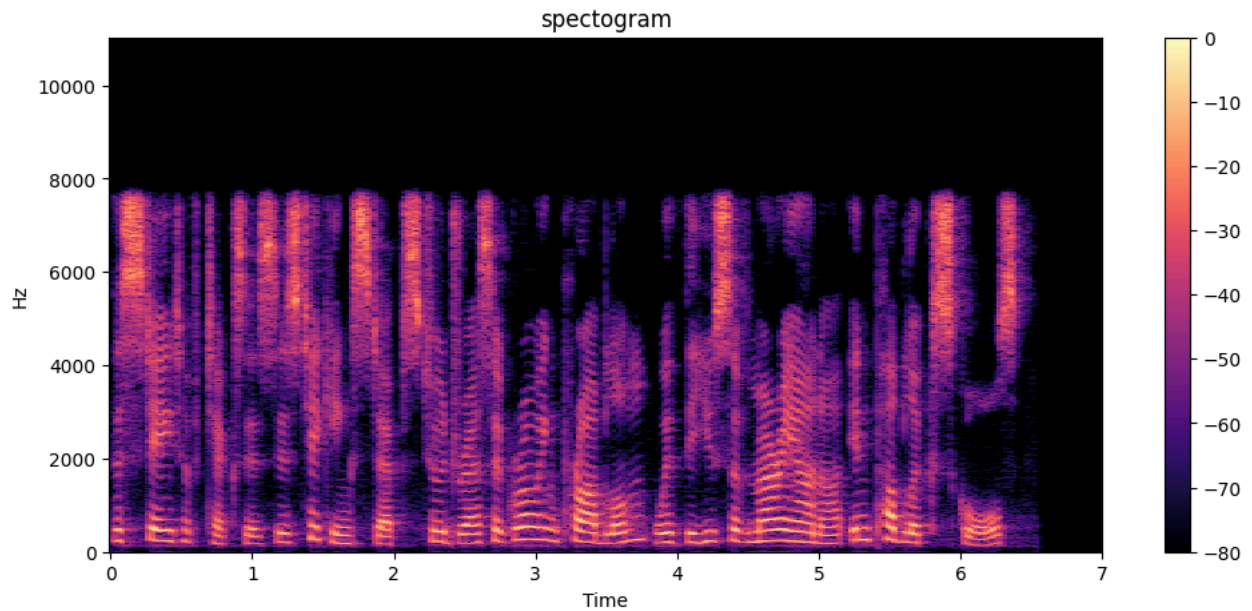
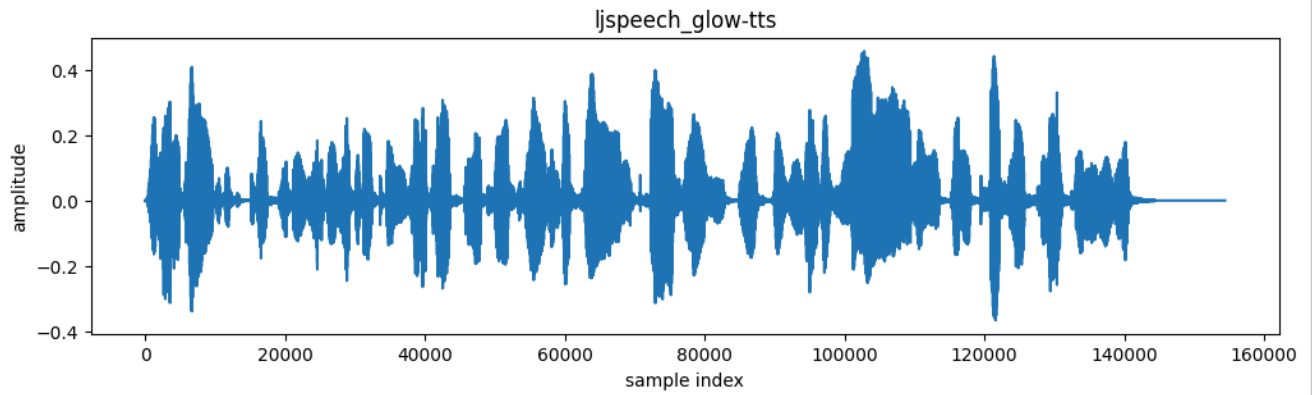
import librosa
y,sr = librosa.load("/content/drive/MyDrive/data/archive/wavs/ljspeech_fastspeech2/7385c5f867ef1df649da476859b8f224.")
print('sampling rate:',sr)
print("duration:",len(y)/sr)
print("shape:",y.shape)
plt.figure(figsize=(12,3))
plt.plot(y)
plt.title(system)
plt.xlabel("sample index")
plt.ylabel("amplitude")
plt.show()
D1 = librosa.amplitude_to_db(np.abs(librosa.stft(y)),ref = np.max)
plt.figure(figsize=(12,5))
librosa.display.specshow(D1,sr=sr,x_axis='time',y_axis='hz')
plt.colorbar()
plt.title("spectrogram")
plt.show()

import librosa
y,sr = librosa.load("/content/drive/MyDrive/data/archive/wavs/ljspeech_glow-tts/b7d53f6c8301665c8c1ae0d647208ecf.wav")
print("sampling rate:",sr)
print("duration:",len(y)/sr)
print("shape:",y.shape)
plt.figure(figsize=(12,3))
plt.plot(y)
plt.title(system)
```

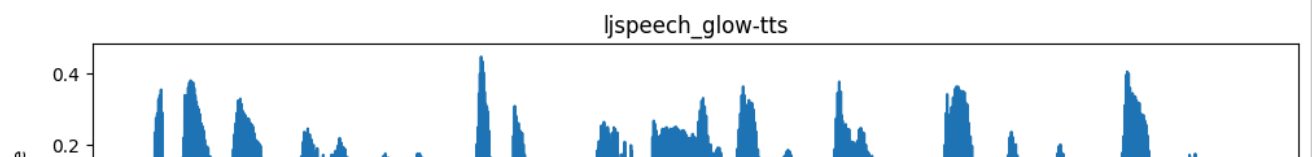
```
plt.xlabel("sample index")
plt.ylabel("amplitude")
plt.show()
D2 = librosa.amplitude_to_db(np.abs(librosa.stft(y)), ref = np.max)
plt.figure(figsize=(12,5))
librosa.display.specshow(D2, sr=sr, x_axis='time', y_axis='hz')
plt.colorbar()
plt.title("spectrogram")
plt.show()
```

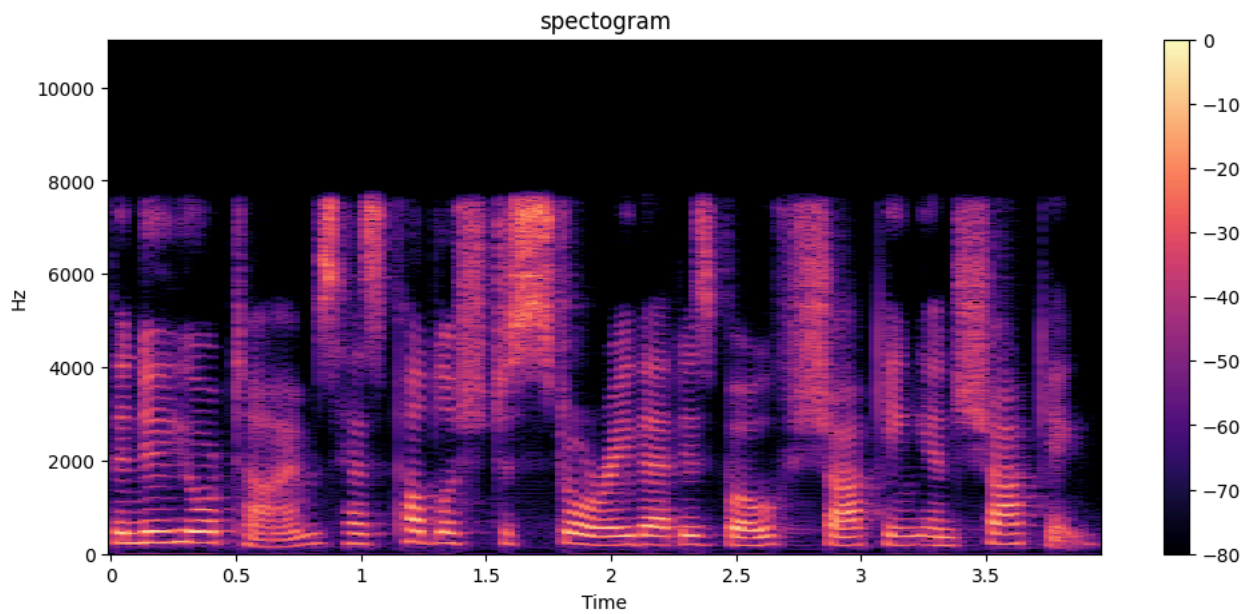
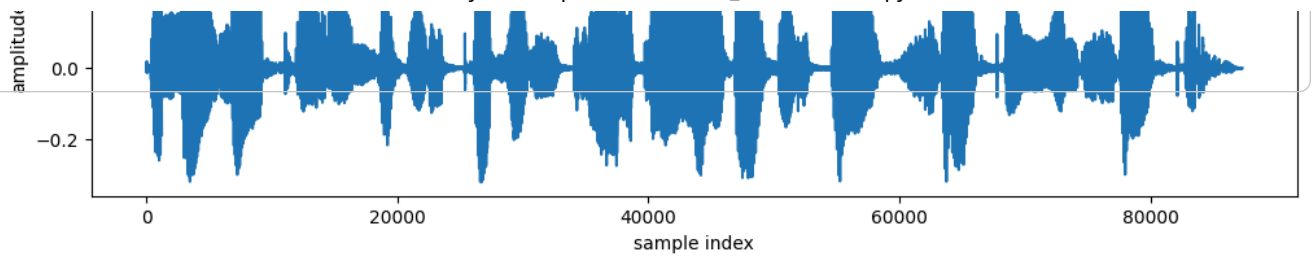


```
sampling rate: 22050  
duration: 7.001587301587302  
shape: (154385,)
```

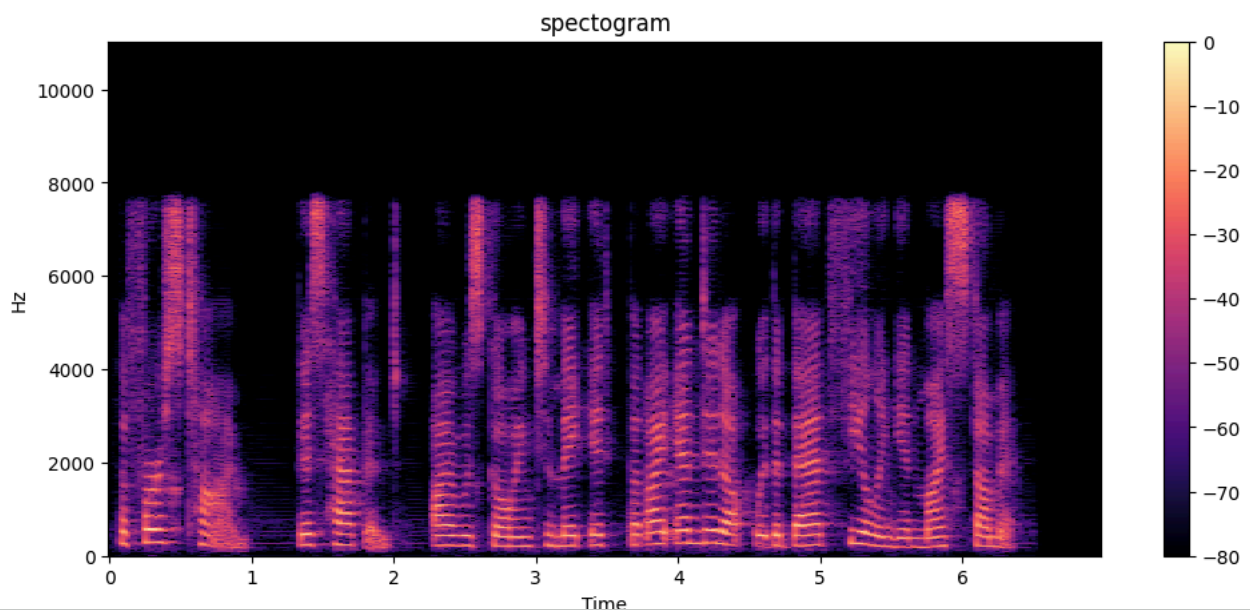
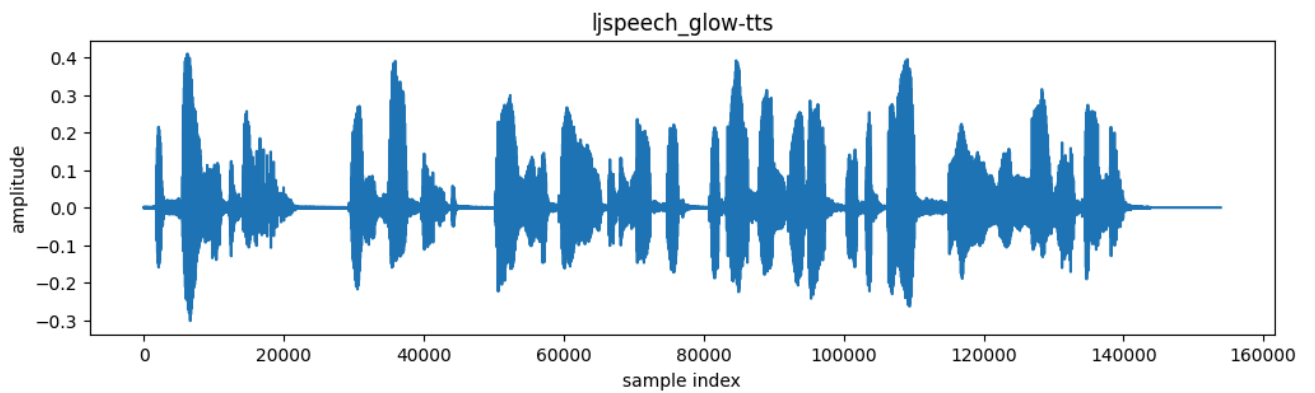


```
sampling rate: 22050  
duration: 3.959092970521542  
shape: (87298,)
```





sampling rate: 22050
duration: 6.978412698412699
shape: (153874,)



```
y, sr = librosa.load("/content/drive/MyDrive/data/archive/wavs/ljspeech_vits/1059df5984efbd52824922d66db379ff.wav")
zcr = librosa.feature.zero_crossing_rate(y)
print(zcr.shape)
print(zcr.mean())

centroid = librosa.feature.spectral_centroid(y=y, sr=sr)
print(centroid.shape)
print("centroid mean:", centroid.mean())

bandwidth = librosa.feature.spectral_bandwidth(y=y, sr=sr)
print(bandwidth.shape)
print("bandwidth mean:", bandwidth.mean())

rolloff = librosa.feature.spectral_rolloff(y=y, sr=sr)
print(rolloff.shape)
print("rolloff mean:", rolloff.mean())

mfcc = librosa.feature.mfcc(y=y, sr=sr, n_mfcc=13)
print(mfcc.shape)
plt.figure(figsize=(10, 4))
librosa.display.specshow(mfcc, x_axis='time')
plt.colorbar()
plt.title('mfcc')
plt.tight_layout()
plt.show()

mfcc_mean = mfcc.mean(axis=1)
mfcc_std = mfcc.std(axis=1)

for i in range(5):
    print(f"MFCC{i+1}:")
    print("Mean = ", mfcc[i].mean())
    print("Std = ", mfcc[i].std())
    print()

rms = librosa.feature.rms(y=y)
print(rms.shape)
print("Mean RMS:", rms.mean())
print("Std RMS:", rms.std())
```